Deliverables for P031

Mandatory Deliverables (70%)

Traffic simulation and data collection. Set up a functioning SUMO traffic simulation with at least one signalised junction. Generate or collect reproducible traffic scenarios, including normal traffic, congestion, and inefficient traffic light configurations. Record relevant traffic measurements such as queue lengths, waiting times, and vehicle travel times.
Due: Autumn trimester

Web-based traffic monitoring platform. Develop a basic web application connected to SUMO through TraCI, enabling users to monitor simulated traffic conditions and traffic light states through an interactive interface.
Due: Autumn trimester

Traffic signal inefficiency detection. Implement a basic rule-based or machine-learning algorithm to identify junctions where inefficient traffic light configurations contribute to congestion. The detection mechanism should distinguish, where possible, signal-related inefficiencies from congestion primarily caused by high traffic demand.
Due: Spring trimester

Traffic signal optimisation and human approval. Implement a basic optimisation mechanism (e.g., rule-based adjustment of green-light durations) to recommend improved signal configurations. Allow a human operator to accept or reject the recommendations before changes are applied in SUMO.
Due: Spring trimester

Integrated system and performance evaluation. Integrate traffic monitoring, inefficiency detection, signal optimisation, and human approval into a complete working system. Quantitatively evaluate its effectiveness against the original traffic signal configurations using metrics such as average waiting time, queue length, and travel time. The dashboard should visualise the effectiveness of approved changes.
Due: Spring trimester

Discretionary Deliverables (20%)

Advanced traffic signal optimisation. Implement more sophisticated optimisation methods (e.g., reinforcement learning, heuristic search, or predictive optimisation) and

compare their effectiveness with the mandatory baseline method.
Due: Spring trimester

Complex traffic scenarios and enhanced human interaction. Extend the system to multiple interconnected junctions, coordinated traffic signal optimisation, or more complex and dynamically varying traffic conditions. Alternatively, implement advanced human-in-the-loop features such as interpretable recommendations, historical decision analysis, or rollback capabilities. Demonstrate measurable improvements beyond the mandatory system.
Due: Spring trimester

Exceptional Deliverables (10%)

Novel optimisation approach or research-level contribution. Propose and validate an original approach to traffic signal inefficiency detection, optimisation, or human-AI collaboration that demonstrates clear advantages over established methods. Support the contribution through rigorous comparative experiments, potentially leading to a conference-quality research paper (e.g., IEEE ITSC). Actual publication is not required.
Due: Spring trimester